

MEHMOOD KHAN

AI ENGINEER | NLP • RAG • GENERATIVE AI • AGENTIC AI

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PROFESSIONAL SUMMARY

BS Computer Science student specializing in Artificial Intelligence, with a focus on Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Generative AI, and Agentic AI. Hands-on experience designing and building AI-powered applications in Python using modern LLM technologies, including document processing, embedding generation, semantic retrieval, and context-aware response generation. Continuously developing practical AI engineering skills through applied project work, with the goal of building intelligent, reliable systems that solve real-world problems.

TECHNICAL SKILLS

Programming: Python

AI & NLP: Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Generative AI, LLM Applications, Embeddings, Semantic Search

AI Frameworks: LangChain, LangGraph

LLM / Vector Technologies: Google Gemini API, Hugging Face Embeddings, Qdrant Cloud

Backend & APIs: FastAPI, REST APIs, PyMuPDF

Frontend: React.js, Tailwind CSS

PROJECTS

Intelligent Research Paper Assistant

Python • FastAPI • React.js • LangChain • RAG • Google Gemini API • Hugging Face Embeddings • Qdrant Cloud • PyMuPDF • REST APIs • Tailwind CSS

- Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) application that allows users to upload research papers and ask natural-language questions about their content.
- Built a document processing pipeline using PyMuPDF to extract and prepare text from PDF research papers for retrieval.
- Generated Hugging Face embeddings and implemented semantic similarity search using Qdrant Cloud as the vector database.
- Integrated Google Gemini to generate grounded, context-aware answers with source citations, and exposed the system through a FastAPI backend with a React.js and Tailwind CSS frontend.

Intelligent Knowledge Assistant

Python • FastAPI • React.js • LangChain • RAG • Google Gemini API • Hugging Face Embeddings • Qdrant Cloud • PyMuPDF • REST APIs • Tailwind CSS

- Developed a document-based knowledge assistant that lets users upload documents and interact with their content through natural-language queries.
- Implemented a RAG pipeline covering document upload, text chunking, and Hugging Face-based embedding generation.
- Configured semantic retrieval with Qdrant Cloud and built context-aware response generation using Google Gemini, served via a FastAPI backend with a React.js and Tailwind CSS interface.

EDUCATION

BS Computer Science

Quaid-e-Awam University of Engineering, Science and Technology, Nawabshah, Sindh, Pakistan
2023 – Expected 2027